

Demographics, Trade, and Growth

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Abstract

To be completed.

Keywords:

JEL Classification:

1 Introduction

To be completed.

2 Model

Time is discrete and extends indefinitely. The world economy consists of N countries, indexed by $i, n, \ell \in \{1, \dots, N\}$. Age groups are indexed by $g \in \mathcal{G} \equiv \{1, \dots, G\}$, and tradable varieties by $\omega \in [0, 1]$. At the beginning of period t , country i is characterized by its population distribution $\{N_{i,g,t}\}_{g \in \mathcal{G}}$, physical-capital stock $K_{i,t}$, and knowledge stock $\lambda_{i,t}$. Fertility, survival, and labor-force participation determine the exogenous demographic path, while physical capital and knowledge evolve endogenously.

Each firm has a production division and an R&D division. The production division uses labor, physical capital, and composite intermediate inputs to produce a continuum of varieties. At the beginning of each period, the best available productivity for each variety follows a Fréchet distribution determined by the country's knowledge stock. Varieties are traded internationally subject to iceberg trade costs. In each destination–variety market, the supplier with the lowest delivered cost serves the market and charges the delivered cost of the next-best supplier, following the Ricardian–Bertrand structure of [Eaton and Kortum \(2002\)](#) and [Bernard et al. \(2003\)](#).

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The R&D division employs researchers from different age groups and generates new productivity points through a Poisson process. Research productivity varies with age, so the contribution of an age group to innovation depends on both its size and its research ability. New productivity points are added to the country's knowledge stock at the end of the period and become available for production in period $t + 1$. This timing separates current production from current research while allowing the value of future operating profits to determine firms' demand for researchers (Kortum, 1997; Eaton and Kortum, 1999, 2007; Santacreu, 2015; Cai et al., 2022). The production and R&D divisions are jointly owned by domestic households, which receive the firm's net payout.

Households live for at most G periods and face age-dependent survival probabilities. Conditional on labor-force participation, an individual of age g in country n supplies $l_{n,g}$ efficiency units of labor and can be assigned to either production or research. Households use labor income, capital income, and firm payouts to finance consumption and saving. Saving is invested in domestic physical capital and determines $K_{i,t+1}$, while research determines $\lambda_{i,t+1}$.

Demographic change affects the economy through three margins. It changes the supply of effective labor through participation and age-specific labor efficiency, the creation of knowledge through the age composition of researchers, and physical-capital accumulation through life-cycle saving. International trade connects these margins across countries by affecting prices, factor returns, market access, operating profits, and the value of new knowledge. The joint evolution of population, capital, and knowledge determines both the transition path and the stationary balanced-growth equilibrium (Ravikumar et al., 2019; Sposi et al., 2021; Sposi, 2022).

2.1 Demographics, participation, and labor allocation

Let $N_{n,g,t}$ denote the number of age- g households alive in country n at calendar date t . Households live for at most G periods. Let $f_{n,g,t}$ denote births per age- g household, and let $s_{n,g,t} \in [0, 1]$ denote the probability of surviving from age $g - 1$ to age g , with survival realized at calendar date t . In particular, $s_{n,1,t}$ is newborn survival and $s_{n,G+1,t} = 0$. Population evolves according to

$$N_{n,1,t+1} = s_{n,1,t} \sum_{g=1}^G f_{n,g,t} N_{n,g,t}, \quad (1)$$

$$N_{n,g+1,t+1} = s_{n,g+1,t+1} N_{n,g,t}, \quad g = 1, \dots, G - 1. \quad (2)$$

Total population and the population share of age group g are

$$N_{n,t} \equiv \sum_{g=1}^G N_{n,g,t}, \quad n_{n,g,t} \equiv \frac{N_{n,g,t}}{N_{n,t}}, \quad (3)$$

with

$$\sum_{g=1}^G n_{n,g,t} = 1. \quad (4)$$

The parameter $l_{n,g} \geq 0$ denotes the efficiency units supplied by one active age- g worker in country n . It captures both the country and age dimensions of effective labor and may combine education and human capital, the working-age profile, hours conditional on participation, and other age-specific productivity differences. The value $l_{n,g} = 0$ outside working ages is allowed. Let $\tau_{n,t}^L \in [0, 1]$ be the fraction of potential workers outside the labor force, so $1 - \tau_{n,t}^L$ is the participation rate. Active age- g labor and its efficiency-unit counterpart are

$$L_{n,g,t} \equiv (1 - \tau_{n,t}^L) N_{n,g,t}, \quad \mathcal{L}_{n,g,t} \equiv l_{n,g} L_{n,g,t}. \quad (5)$$

Conditional on participation, firms assign a fraction $s_{n,g,t}^R \in [0, 1]$ of active age- g workers to research. Researcher and production-worker headcounts are

$$L_{n,g,t}^R \equiv s_{n,g,t}^R L_{n,g,t}, \quad L_{n,g,t}^P \equiv (1 - s_{n,g,t}^R) L_{n,g,t}, \quad (6)$$

and aggregate efficiency labor in the two activities is

$$\mathcal{L}_{n,t}^R \equiv \sum_{g=1}^G l_{n,g} L_{n,g,t}^R, \quad \mathcal{L}_{n,t}^P \equiv \sum_{g=1}^G l_{n,g} L_{n,g,t}^P. \quad (7)$$

The competitive wage $W_{n,t}$ is paid per efficiency unit. An active age- g worker therefore earns $W_{n,t} l_{n,g}$, whether assigned to production or research. The representative age- g household receives average labor income

$$W_{n,t} (1 - \tau_{n,t}^L) l_{n,g}. \quad (8)$$

Thus, the production–research assignment does not alter the household’s labor income or its individual saving problem.

2.2 Production, trade, and operating profits

Country n combines a continuum of varieties into a nontradable composite good,

$$Q_{n,t} = \left[\int_0^1 x_{n,t}(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right]^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1, \quad (9)$$

where $x_{n,t}(\omega)$ is total demand for variety ω . Let $P_{n,t}$ be the corresponding ideal price index and $X_{n,t} \equiv P_{n,t} Q_{n,t}$ nominal expenditure.

A producer in origin i with productivity $q_{i,t}(\omega)$ uses efficiency labor $h_{i,t}(\omega)$, physical capital $k_{i,t}(\omega)$, and the composite intermediate input $m_{i,t}(\omega)$:

$$y_{i,t}(\omega) = q_{i,t}(\omega) \left[h_{i,t}(\omega)^{\gamma_{L,i}} k_{i,t}(\omega)^{1-\gamma_{L,i}} \right]^{\gamma_i} m_{i,t}(\omega)^{1-\gamma_i}. \quad (10)$$

Let $R_{i,t}$ be the rental payment per unit of capital. Unit input cost before dividing by productivity is

$$c_{i,t} = Y_i \left(W_{i,t}^{\gamma_{L,i}} R_{i,t}^{1-\gamma_{L,i}} \right)^{\gamma_i} P_{i,t}^{1-\gamma_i}, \quad (11)$$

where Y_i is the Cobb–Douglas cost constant defined in Appendix A.3. Shipping one unit from origin i to destination n requires $\kappa_{ni,t} \geq 1$ units at origin, so delivered unit cost is $\kappa_{ni,t} c_{i,t} / q$. Bilateral variables use destination–origin ordering throughout.

The best productivity available for a variety in country i is Fréchet:

$$\Pr[q_{i,t}(\omega) \leq q] = \exp[-\lambda_{i,t} q^{-\theta}], \quad \theta > 0. \quad (12)$$

Define destination n 's competitiveness index

$$\Phi_{n,t} \equiv \sum_{\ell=1}^N \lambda_{\ell,t} (c_{\ell,t} \kappa_{n\ell,t})^{-\theta}. \quad (13)$$

The lowest-cost supplier serves the market and charges the second-lowest delivered cost. The resulting bilateral expenditure share and price index are

$$\pi_{ni,t} \equiv \frac{X_{ni,t}}{X_{n,t}} = \frac{\lambda_{i,t} (c_{i,t} \kappa_{ni,t})^{-\theta}}{\Phi_{n,t}}, \quad (14)$$

$$P_{n,t} = \mathcal{B}(\theta, \sigma) \Phi_{n,t}^{-1/\theta}, \quad \mathcal{B}(\theta, \sigma) \equiv \Gamma\left(2 - \frac{\sigma-1}{\theta}\right)^{\frac{1}{1-\sigma}}. \quad (15)$$

We impose $\theta > \sigma - 1$.

Country i 's gross sales and operating profits are

$$Y_{i,t} \equiv \sum_{n=1}^N \pi_{ni,t} X_{n,t}, \quad \Pi_{i,t}^{\text{op}} = \frac{1}{1+\theta} Y_{i,t}. \quad (16)$$

Factor payments therefore satisfy

$$W_{i,t} \mathcal{L}_{i,t}^P = \gamma_i \gamma_{L,i} \frac{\theta}{1+\theta} Y_{i,t}, \quad (17)$$

$$R_{i,t} K_{i,t} = \gamma_i (1 - \gamma_{L,i}) \frac{\theta}{1+\theta} Y_{i,t}, \quad (18)$$

$$P_{i,t} M_{i,t} = (1 - \gamma_i) \frac{\theta}{1+\theta} Y_{i,t}. \quad (19)$$

Only $\mathcal{L}_{i,t}^P$ enters current production; research labor affects output through future knowledge.

2.3 Research assignment and knowledge accumulation

Country i 's current research productivity depends on the inherited knowledge stock:

$$\alpha_{i,t} = \bar{\alpha}_i \lambda_{i,t}^\nu, \quad \bar{\alpha}_i > 0, \quad 0 \leq \nu < 1. \quad (20)$$

Within the active workforce of age g , workers are ordered by $u \in (0, 1]$. A rank- u worker contributes

$$b_{i,g,t}(u) = \alpha_{i,t} \eta_g \xi u^{\xi-1}, \quad \eta_g > 0, \quad 0 < \xi < 1, \quad (21)$$

to the Poisson arrival intensity of new ideas.

If a fraction $s_{i,g,t}^R$ of active age- g workers is assigned to research, the knowledge increment generated by this age group is

$$\begin{aligned} \Delta \lambda_{i,g,t} &= L_{i,g,t} \int_0^{s_{i,g,t}^R} b_{i,g,t}(u) du \\ &= \bar{\alpha}_i \lambda_{i,t}^\nu \eta_g \left(s_{i,g,t}^R \right)^\xi L_{i,g,t}. \end{aligned} \quad (22)$$

Total new knowledge is

$$\Delta \lambda_{i,t} = \bar{\alpha}_i \lambda_{i,t}^\nu \sum_{g=1}^G \eta_g \left(s_{i,g,t}^R \right)^\xi L_{i,g,t}, \quad (23)$$

and

$$\lambda_{i,t+1} = \lambda_{i,t} + \Delta \lambda_{i,t}. \quad (24)$$

Let $v_{i,t}$ be the date- t nominal value of one unit of knowledge created during period t . The

age- g research problem is

$$\max_{0 \leq s \leq 1} L_{i,g,t} \left[v_{i,t} \bar{\alpha}_i \lambda_{i,t}^v \eta_g s^{\bar{\zeta}} - W_{i,t} l_{i,g} s \right]. \quad (25)$$

For an interior assignment,

$$\bar{\zeta} v_{i,t} \bar{\alpha}_i \lambda_{i,t}^v \eta_g \left(s_{i,g,t}^R \right)^{\bar{\zeta}-1} = W_{i,t} l_{i,g}. \quad (26)$$

Hence

$$s_{i,g,t}^R = \min \left\{ 1, \left(\frac{\bar{\zeta} v_{i,t} \bar{\alpha}_i \lambda_{i,t}^v \eta_g}{W_{i,t} l_{i,g}} \right)^{\frac{1}{1-\bar{\zeta}}} \right\}, \quad l_{i,g} > 0. \quad (27)$$

The participation factor $1 - \tau_{i,t}^L$ cancels from the private first-order condition but affects research assignment indirectly through general equilibrium.

2.4 OLG households and endogenous physical capital

A household born at calendar date t is age g at date $t + g - 1$. Its consumption and asset choices are

$$\{c_{n,g,t+g-1}\}_{g=1}^G, \quad \{a_{n,g+1,t+g}\}_{g=1}^{G-1}.$$

Period utility is CRRA:

$$u(c) = \begin{cases} \frac{c^{1-1/\rho}}{1-1/\rho}, & \rho \neq 1, \\ \log c, & \rho = 1, \end{cases} \quad \rho > 0. \quad (28)$$

Let $\beta \in (0, 1)$ be the discount factor and $\psi_{n,\tau} > 0$ the saving wedge associated with assets that pay off at calendar date τ . Define survival to age g by

$$S_{n,g,t+g-1} \equiv \prod_{h=2}^g s_{n,h,t+h-1}, \quad S_{n,1,t} = 1. \quad (29)$$

The household maximizes

$$\sum_{g=1}^G \beta^{g-1} S_{n,g,t+g-1} u(c_{n,g,t+g-1}). \quad (30)$$

Let $D_{n,\tau}^F$ denote the aggregate net payout of the consolidated domestic corporate sector and $d_{n,\tau}^F \equiv D_{n,\tau}^F / N_{n,\tau}$ its per-capita distribution. Let $tr_{n,\tau}^D$ denote the accidental-bequest trans-

fer and $tr_{n,\tau}^T$ the international transfer. At age g ,

$$\begin{aligned} P_{n,t+g-1}c_{n,g,t+g-1} + P_{n,t+g-1}a_{n,g+1,t+g} &= P_{n,t+g-1}(1 + r_{n,t+g-1})a_{n,g,t+g-1} \\ &+ W_{n,t+g-1}(1 - \tau_{n,t+g-1}^L)l_{n,g} \\ &+ d_{n,t+g-1}^F + tr_{n,t+g-1}^D + tr_{n,t+g-1}^T. \end{aligned} \quad (31)$$

Boundary conditions are

$$a_{n,1,t} = 0, \quad a_{n,G+1,t+G} = 0, \quad c_{n,g,t+g-1} > 0. \quad (32)$$

The net real return is

$$r_{n,\tau} \equiv \frac{R_{n,\tau}}{P_{n,\tau}} - \delta. \quad (33)$$

For an unconstrained saver,

$$u'(c_{n,g,t+g-1}) = \beta \psi_{n,t+g} s_{n,g+1,t+g} (1 + r_{n,t+g}) u'(c_{n,g+1,t+g}), \quad g = 1, \dots, G-1. \quad (34)$$

Aggregate consumption and capital are

$$C_{n,t} \equiv \sum_{g=1}^G N_{n,g,t} c_{n,g,t}, \quad (35)$$

$$K_{n,t} \equiv \sum_{g=2}^G N_{n,g-1,t-1} a_{n,g,t}, \quad (36)$$

$$K_{n,t+1} \equiv \sum_{g=1}^{G-1} N_{n,g,t} a_{n,g+1,t+1}. \quad (37)$$

Gross investment is

$$I_{n,t} \equiv K_{n,t+1} - (1 - \delta)K_{n,t}. \quad (38)$$

Capital is internationally immobile in the baseline.

2.5 Corporate payouts and the value of knowledge

Variety production generates $\Pi_{i,t}^{\text{op}}$, while innovation firms pay the research wage bill $W_{i,t}\mathcal{L}_{i,t}^R$. The consolidated corporate payout is

$$D_{i,t}^F = \Pi_{i,t}^{\text{op}} - W_{i,t}\mathcal{L}_{i,t}^R. \quad (39)$$

A unit of knowledge created during t first earns operating profits in $t + 1$. No-arbitrage gives

$$v_{i,t} = \frac{P_{i,t}}{P_{i,t+1}(1 + r_{i,t+1})} \left[\frac{\Pi_{i,t+1}^{\text{op}}}{\lambda_{i,t+1}} + v_{i,t+1} \right]. \quad (40)$$

2.6 Aggregation and equilibrium dynamics

Let $D_{i,t}$ be country i 's trade deficit, with $\sum_i D_{i,t} = 0$, and set $tr_{i,t}^T = D_{i,t}/N_{i,t}$. Aggregating household budgets and redistributing accidental bequests yields

$$\begin{aligned} P_{i,t}C_{i,t} + P_{i,t}I_{i,t} &= W_{i,t}\mathcal{L}_{i,t}^P + W_{i,t}\mathcal{L}_{i,t}^R + R_{i,t}K_{i,t} \\ &\quad + D_{i,t}^F + D_{i,t}. \end{aligned} \quad (41)$$

Substituting (39) gives

$$P_{i,t}C_{i,t} + P_{i,t}I_{i,t} = W_{i,t}\mathcal{L}_{i,t}^P + R_{i,t}K_{i,t} + \Pi_{i,t}^{\text{op}} + D_{i,t}. \quad (42)$$

The composite-good resource constraint is

$$Q_{i,t} = C_{i,t} + I_{i,t} + M_{i,t}, \quad X_{i,t} = P_{i,t}Q_{i,t} = Y_{i,t} + D_{i,t}. \quad (43)$$

Definition 1 (Transitional equilibrium) *Given initial knowledge and age-specific assets, $\{\lambda_{n,0}, a_{n,g,0}\}_{n,g}$, together with demographic paths, participation paths, trade costs, trade-deficit paths, preference shifters, and model parameters, a transitional equilibrium is a sequence of demographic allocations, household choices, research assignments, capital and knowledge stocks, prices, trade flows, and corporate payouts satisfying household optimality; innovation-firm optimality; the demographic, capital, and knowledge laws of motion; production and trade; labor, capital, and goods-market clearing; corporate and aggregate accounting; one nominal normalization each period; and the knowledge-value no-bubble condition.*

2.7 Balanced growth and the stationary steady state

Suppose that, after date T , all countries share a constant population growth rate g_n , stable age shares, constant survival and fertility rates, constant participation, stationary research-assignment shares, constant trade costs and wedges, and a common gross-output labor share

$$\alpha_L \equiv \gamma_i \gamma_{L,i}. \quad (44)$$

Proposition 1 (Balanced-growth rates) *On a balanced growth path with interior stationary research-*

assignment shares,

$$1 + g_\lambda = (1 + g_n)^{\frac{1}{1-\nu}}, \quad (45)$$

$$1 + g_A = (1 + g_\lambda)^{1/\theta}, \quad (46)$$

$$1 + g_\omega = (1 + g_A)^{1/\alpha_L} = (1 + g_n)^{\frac{1}{\theta\alpha_L(1-\nu)}}, \quad (47)$$

$$1 + g_K = (1 + g_n)(1 + g_\omega). \quad (48)$$

The net real capital return and bilateral trade shares are stationary. If $g_n = 0$, then $g_\lambda = g_A = g_\omega = g_K = 0$.

3 Data and Calibration

To be completed.

4 Quantitative Results

To be completed.

5 Counterfactuals

To be completed.

6 Conclusion

To be completed.

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A Model Details and Derivations

This appendix collects the detailed demographic, household, production, trade, innovation, valuation, accounting, balanced-growth, and numerical equilibrium systems underlying Section 2. The main text keeps only the equations needed to understand the economic mechanism.

A.1 Demography and labor-force participation

Let

$$\mathbf{N}_{n,t} \equiv (N_{n,1,t}, \dots, N_{n,G,t})'.$$

The demographic laws can be written as

$$\mathbf{N}_{n,t+1} = \mathbf{\Pi}_{n,t} \mathbf{N}_{n,t}. \quad (49)$$

If demographic rates become constant, define

$$\mathbf{n}_n \equiv (n_{n,1}, \dots, n_{n,G})', \quad (50)$$

with

$$\mathbf{\Pi}_n \mathbf{n}_n = (1 + g_n) \mathbf{n}_n, \quad \mathbf{1}' \mathbf{n}_n = 1. \quad (51)$$

The country–age efficiency profile $l_{n,g}$ is primitive, while participation is kept separate:

$$L_{n,g,t} = (1 - \tau_{n,t}^L) N_{n,g,t}, \quad \mathcal{L}_{n,g,t} = l_{n,g} (1 - \tau_{n,t}^L) N_{n,g,t}. \quad (52)$$

A.2 Financial market, household first-order conditions, and bequests

Because the baseline is financially autarkic,

$$K_{n,t} = \sum_{g=2}^G N_{n,g-1,t-1} a_{n,g,t}. \quad (53)$$

Market clearing implies

$$r_{n,t} = \frac{R_{n,t}}{P_{n,t}} - \delta. \quad (54)$$

Let $\mu_{n,g,\tau}$ be the multiplier on the household budget constraint and $\chi_{n,g,\tau} \geq 0$ the multiplier on $a_{n,g+1,\tau+1} \geq 0$. Then

$$u'(c_{n,g,\tau}) = \mu_{n,g,\tau} P_{n,\tau}, \quad (55)$$

$$\mu_{n,g,\tau} P_{n,\tau} = \beta \psi_{n,\tau+1} s_{n,g+1,\tau+1} \mu_{n,g+1,\tau+1} P_{n,\tau+1} (1 + r_{n,\tau+1}) + \chi_{n,g,\tau}, \quad (56)$$

$$\chi_{n,g,\tau} a_{n,g+1,\tau+1} = 0. \quad (57)$$

Under CRRA preferences, an unconstrained saver satisfies

$$\frac{c_{n,g+1,t+g}}{c_{n,g,t+g-1}} = [\beta \psi_{n,t+g} s_{n,g+1,t+g} (1 + r_{n,t+g})]^\rho. \quad (58)$$

Claims left by households that do not survive are

$$K_{n,t}^B \equiv \sum_{g=2}^G (N_{n,g-1,t-1} - N_{n,g,t}) a_{n,g,t}. \quad (59)$$

Their date- t payoff is

$$T_{n,t}^B = P_{n,t}(1 + r_{n,t})K_{n,t}^B = [R_{n,t} + (1 - \delta)P_{n,t}] K_{n,t}^B. \quad (60)$$

Under equal-per-capita redistribution,

$$tr_{n,t}^D = \frac{T_{n,t}^B}{N_{n,t}}. \quad (61)$$

The international transfer is

$$tr_{n,t}^T = \frac{D_{n,t}}{N_{n,t}}, \quad \sum_{n=1}^N D_{n,t} = 0. \quad (62)$$

A.3 Composite demand and production cost

Expenditure minimization gives

$$P_{n,t} = \left[\int_0^1 p_{n,t}(\omega)^{1-\sigma} d\omega \right]^{\frac{1}{1-\sigma}}, \quad (63)$$

$$x_{n,t}(\omega) = \left(\frac{p_{n,t}(\omega)}{P_{n,t}} \right)^{-\sigma} Q_{n,t}. \quad (64)$$

Define

$$Y_i \equiv (\gamma_i \gamma_{L,i})^{-\gamma_i \gamma_{L,i}} [\gamma_i (1 - \gamma_{L,i})]^{-\gamma_i (1 - \gamma_{L,i})} (1 - \gamma_i)^{-(1 - \gamma_i)}. \quad (65)$$

Then

$$c_{i,t} = Y_i \left(W_{i,t}^{\gamma_{L,i}} R_{i,t}^{1 - \gamma_{L,i}} \right)^{\gamma_i} P_{i,t}^{1 - \gamma_i}. \quad (66)$$

A.4 Pareto–Poisson foundation of the knowledge frontier

Introduce a temporary quality lower bound $\bar{z} > 0$. Conditional idea quality has Pareto CDF

$$H_{\bar{z}}(z) = \begin{cases} 0, & z < \bar{z}, \\ 1 - (z/\bar{z})^{-\theta}, & z \geq \bar{z}. \end{cases} \quad (67)$$

For each variety, the number of new ideas generated during t is

$$\mathcal{N}_{i,t}^{\text{new},\bar{z}}(\omega) \sim \text{Poisson}\left(\Delta\lambda_{i,t}\bar{z}^{-\theta}\right). \quad (68)$$

For $q \geq \bar{z}$,

$$F_{i,t}^{\text{best,new}}(q;\bar{z}) = \exp[-\Delta\lambda_{i,t}q^{-\theta}]. \quad (69)$$

Taking $\bar{z} \downarrow 0$ and superimposing inherited and new processes gives

$$\lambda_{i,t+1} = \lambda_{i,t} + \Delta\lambda_{i,t}, \quad F_{i,t}(q) = \exp[-\lambda_{i,t}q^{-\theta}]. \quad (70)$$

A.5 Trade derivations and factor payments

Define

$$\varphi_{ni,t} \equiv \lambda_{i,t}(c_{i,t}\kappa_{ni,t})^{-\theta}, \quad \Phi_{n,t} \equiv \sum_{\ell} \varphi_{n\ell,t}. \quad (71)$$

The supplier probability equals the expenditure share:

$$\pi_{ni,t} = \frac{\lambda_{i,t}(c_{i,t}\kappa_{ni,t})^{-\theta}}{\sum_{\ell} \lambda_{\ell,t}(c_{\ell,t}\kappa_{n\ell,t})^{-\theta}}. \quad (72)$$

The second-order statistic of the delivered-cost process gives

$$P_{n,t} = \Gamma\left(2 - \frac{\sigma - 1}{\theta}\right)^{\frac{1}{1-\sigma}} \Phi_{n,t}^{-1/\theta}, \quad (73)$$

and the expected variable-cost share is $\theta/(1 + \theta)$, implying

$$\Pi_{i,t}^{\text{op}} = \frac{1}{1 + \theta} Y_{i,t}. \quad (74)$$

Factor payments are therefore

$$W_{i,t}\mathcal{L}_{i,t}^P = \gamma_i\gamma_{L,i}\frac{\theta}{1 + \theta}Y_{i,t}, \quad (75)$$

$$R_{i,t}K_{i,t} = \gamma_i(1 - \gamma_{L,i})\frac{\theta}{1 + \theta}Y_{i,t}, \quad (76)$$

$$P_{i,t}M_{i,t} = (1 - \gamma_i)\frac{\theta}{1 + \theta}Y_{i,t}. \quad (77)$$

A.6 Innovation firms and the research cutoff

Let $j \in [0, 1]$ index atomistic innovation firms. Firm j chooses $\tilde{s}_{i,g,t}^R(j) \in [0, 1]$ to maximize

$$\mathcal{S}_{i,g,t}^R(j, \tilde{s}^R) = L_{i,g,t}(j) \left[v_{i,t} \alpha_{i,t} \eta_g(\tilde{s}^R)^\xi - W_{i,t} l_{i,g} \tilde{s}^R \right]. \quad (78)$$

The interior first-order condition is

$$\xi v_{i,t} \bar{\alpha}_i \lambda_{i,t}^\nu \eta_g[\tilde{s}_{i,g,t}^{R,*}(j)]^{\xi-1} = W_{i,t} l_{i,g}. \quad (79)$$

Using $L_{i,g,t}(j) = (1 - \tau_{i,t}^L) N_{i,g,t}(j)$ shows that the common participation factor cancels from the private first-order condition:

$$\left. \frac{\partial \mathcal{S}_{i,g,t}^R}{\partial \tau_{i,t}^L} \right|_{v_{i,t}, W_{i,t}, \lambda_{i,t}} = 0. \quad (80)$$

In general equilibrium, however,

$$\frac{d\mathcal{S}_{i,g,t}^{R,*}}{d\tau_{i,t}^L} \neq 0. \quad (81)$$

A.7 Corporate accounting and the value of knowledge

Let $\mathcal{T}_{i,t}^K$ be the royalty or licensing payment from variety producers to innovation firms. Then

$$D_{i,t}^P = \Pi_{i,t}^{\text{op}} - \mathcal{T}_{i,t}^K, \quad (82)$$

$$D_{i,t}^R = \mathcal{T}_{i,t}^K - W_{i,t} \mathcal{L}_{i,t}^R, \quad (83)$$

so the consolidated payout is

$$D_{i,t}^F = \Pi_{i,t}^{\text{op}} - W_{i,t} \mathcal{L}_{i,t}^R. \quad (84)$$

Treat $\lambda_{i,0}$ as inherited knowledge. Research conducted during period r creates a cohort of size $\Delta \lambda_{i,r}$, so

$$\lambda_{i,\tau} = \lambda_{i,0} + \sum_{r=0}^{\tau-1} \Delta \lambda_{i,r}. \quad (85)$$

The expected flow payoff per unit of knowledge at date τ is

$$\frac{\Pi_{i,\tau}^{\text{op}}}{\lambda_{i,\tau}}. \quad (86)$$

The date- t stochastic discount factor to $\tau > t$ is

$$m_{i,t,\tau}^K = \frac{P_{i,t}}{P_{i,\tau}} \prod_{u=t+1}^{\tau} (1 + r_{i,u})^{-1}. \quad (87)$$

Hence

$$v_{i,t} = \sum_{\tau=t+1}^{\infty} m_{i,t,\tau}^K \frac{\Pi_{i,\tau}^{\text{op}}}{\lambda_{i,\tau}}, \quad (88)$$

with no-bubble condition

$$\lim_{T \rightarrow \infty} m_{i,t,T}^K v_{i,T} = 0. \quad (89)$$

A.8 Aggregate accounting

Summing individual budgets gives

$$\begin{aligned} P_{i,t}C_{i,t} + P_{i,t}I_{i,t} &= W_{i,t}\mathcal{L}_{i,t}^P + W_{i,t}\mathcal{L}_{i,t}^R + R_{i,t}K_{i,t} \\ &\quad + D_{i,t}^F + D_{i,t}. \end{aligned} \quad (90)$$

Using $D_{i,t}^F = \Pi_{i,t}^{\text{op}} - W_{i,t}\mathcal{L}_{i,t}^R$ gives

$$P_{i,t}C_{i,t} + P_{i,t}I_{i,t} = W_{i,t}\mathcal{L}_{i,t}^P + R_{i,t}K_{i,t} + \Pi_{i,t}^{\text{op}} + D_{i,t}. \quad (91)$$

Production accounting is

$$Y_{i,t} = W_{i,t}\mathcal{L}_{i,t}^P + R_{i,t}K_{i,t} + P_{i,t}M_{i,t} + \Pi_{i,t}^{\text{op}}. \quad (92)$$

Combining these identities gives

$$P_{i,t}(C_{i,t} + I_{i,t} + M_{i,t}) = Y_{i,t} + D_{i,t} = X_{i,t}. \quad (93)$$

A.9 Complete transitional equilibrium

Given initial conditions $\{\lambda_{i,0}, \{a_{i,g,0}\}_{g=1}^G\}_{i=1}^N$, exogenous demographic and participation paths, trade costs, trade deficits, preference shifters, and parameters, a transitional equilibrium consists of sequences satisfying:

- (i) demography and participation;
- (ii) production–research labor allocation;
- (iii) household optimality and asset boundary conditions;

- (iv) physical-capital accumulation;
- (v) production, trade, factor-market clearing, and goods-market clearing;
- (vi) innovation, knowledge accumulation, and research-assignment optimality;
- (vii) corporate valuation and payouts;
- (viii) accidental-bequest and trade-transfer clearing;
- (ix) one nominal normalization each period; and
- (x) the knowledge-value no-bubble condition.

A.10 Balanced growth and stationary equilibrium

Under stable demographics and participation,

$$L_{n,g,t} = (1 + g_n)^{t-T} L_{n,g,T}, \quad t \geq T. \quad (94)$$

A constant knowledge growth rate requires $L_{n,t}/\lambda_{n,t}^{1-\nu}$ to be stationary. Therefore,

$$(1 + g_\lambda)^{1-\nu} = 1 + g_n, \quad 1 + g_\lambda = (1 + g_n)^{1/(1-\nu)}. \quad (95)$$

A fixed quantile of the Fréchet frontier is proportional to $\lambda_{n,t}^{1/\theta}$, so

$$1 + g_A = (1 + g_\lambda)^{1/\theta}. \quad (96)$$

Let g_ω be the growth rate of real per-capita quantities and g_K the common growth rate of aggregate capital, intermediates, output, and real expenditure. Along a balanced path,

$$1 + g_K = (1 + g_n)(1 + g_\omega), \quad (97)$$

and

$$1 + g_\omega = (1 + g_A)^{1/\alpha_L} = (1 + g_n)^{1/[\theta\alpha_L(1-\nu)]}. \quad (98)$$

If $g_n = 0$, then $g_\lambda = g_A = g_\omega = g_K = 0$.

B Computational Algorithm

B.1 Steady state

Algorithm 1 Open Economy Steady State: Single-Sector Model

- 1: Guess a vector of capital stocks $\mathbf{K}^i = \{K_1, \dots, K_N\}^i$ and calculate $I_n = (g_K + \delta)K_n$.
- 2: Given $\mathbf{K}^i$, solve the production, trade, innovation, and valuation blocks under the chosen nominal normalization.
- 3: Solve the $G - 1$ Euler equations for each country together with age-specific budget constraints and asset boundaries.
- 4: Compute

$$K_n^{i'} = \sum_{g=2}^G \frac{N_{n,g-1,T}}{1 + g_n} a_{n,g}^i.$$

- 5: If $|\mathbf{K}^i - \mathbf{K}^{i'}| < \epsilon$, stop.
- 6: Otherwise update

$$\mathbf{K}^{i+1} = \zeta \mathbf{K}^{i'} + (1 - \zeta) \mathbf{K}^i, \quad \zeta \in (0, 1),$$

and return to Step 2.

B.2 Transition path

A finite-horizon implementation uses a terminal balanced growth path at $T + 1$.

1. Solve the terminal SS/BGP system.
2. Guess paths for capital, research-assignment shares, accidental-bequest transfers, and any endogenous transfer closure.
3. Update knowledge forward from $\lambda_{i,0}$ using (24).
4. At each date, solve the static trade equilibrium taking $(K, \lambda, \mathcal{L}^P)$ as given.
5. Starting from the terminal continuation value, iterate (40) backward.
6. Update research-assignment shares using (27).
7. Solve household problems and aggregate saving to obtain the implied capital path.
8. Recompute accidental bequests and corporate payouts.
9. Dampen and update until all equilibrium residuals are below tolerance.

C Calibration of Saving Wedges

C.1 Steady-state saving wedge

The saving wedge is calibrated so that the model reproduces the observed steady-state capital stock.

1. Guess $\boldsymbol{\psi}^{(j)} = \{\psi_1^{(j)}, \dots, \psi_N^{(j)}\}$.
2. Given prices, wages, returns, transfers, capital, knowledge, and the trial wedges, solve the household block.

3. Compute

$$\tilde{K}_n^{(j)} = \sum_{g=2}^G \frac{N_{n,g-1}}{1 + g_n} a_{n,g}^{(j)}. \quad (99)$$

4. Update the wedge until $\tilde{\mathbf{K}}^{(j)}$ matches $\mathbf{K}^{\text{data}}$ to tolerance.
5. Conditional on the calibrated wedges, solve the remaining steady-state conditions.

C.2 Transition-path saving wedges

The transition saving wedges $\{\psi_{n,t}\}$ are calibrated to match the observed path of aggregate capital, with $\psi_{n,1}$ fixed at its calibrated steady-state value.

1. Guess $\boldsymbol{\psi}_t^{(j)} = \{\psi_{1,t}^{(j)}, \dots, \psi_{N,t}^{(j)}\}$ for $t = 2, \dots, T + 1$.
2. Given the paths of prices, wages, returns, payouts, transfers, survival probabilities, and trial saving wedges, solve the household block.
3. Aggregate the implied household assets:

$$\tilde{K}_{n,t}^{(j)} = \sum_{g=2}^G N_{n,g-1,t-1} a_{n,g,t}^{(j)}. \quad (100)$$

4. Update the wedge path until the implied capital path matches the observed capital path to tolerance.